MSEQ 8

8-band parametric equalizer with Mid/Side processing
with automatic, psychoacoustically-informed resonance detection

User Guide

Version 15

1. What is MSEQ 8?

MSEQ 8 is a parametric equalizer with eight bell bands, high-pass and low-pass filters, and full Mid/Side processing. Every filter can be applied to the mid channel (what left and right have in common - the centre), the side channel (the difference between left and right - the width), or both. This lets you, for example, clean up muddy bass in the stereo width without touching the kick in the centre, or add air to the sides without making the lead vocal harsh.

Its resonance detection and Find Resonances assistant use psychoacoustically-informed resonance detection - they judge peaks by how the ear actually perceives them, not just by raw signal level. See chapter 10 for how it works.

Mid/Side in short

The plugin encodes the stereo signal as mid = (L+R)/2 and side = (L-R)/2, processes the channels separately, and decodes back to left/right (L = mid+side, R = mid-side). A mono signal carries no side energy at all - a band in S mode will then do nothing audible.

> *TIP: If you cannot hear any effect, try M or MS mode first - the signal may be (nearly) mono.*

2. Interface overview

Section	Contents
Header	Preset menu, A/B/C/D snapshots, MONITOR selector (ST/M/S), DELTA listen button, output GAIN knob, BYPASS button.
Graph	Mid and side frequency response curves, spectrum analyser, draggable nodes for bands 1-8 plus HP/LP, legend with display toggles, axis labels.
Meter panel	Large IN/OUT peak meters with a dB scale (0 to -60 dB), to the right of the graph.
Band columns	One column per band: M/MS/S selector, Freq/Gain/Q knobs and a bypass button (0).
Footer	Quick reference for mouse and keyboard interactions.

The window can be resized by dragging the bottom-right corner (1020x550 up to 1800x1100 - the minimum width was raised from 900 px so the header controls no longer crowd together). The size is stored with your project. While resizing, spectrum rendering is briefly paused - audio is never affected. With several instances open at once, an unfocused plugin window automatically throttles its rendering rate to keep the rest of your session responsive; the audio itself is never affected.

3. The EQ bands

Each band is a bell (peak) filter with three parameters: Freq (20 Hz-20 kHz), Gain (+-18 dB) and Q (0.1-10; higher Q = narrower band). The default frequencies are 60, 150, 400, 800, 1600, 3200, 6500 and 12000 Hz.

The M / MS / S selector

Mode	Effect	Node colour
M	The band only affects the mid channel (centre/mono content).	Green
MS	The band affects both mid and side - like a regular stereo EQ. Default.	Purple-grey
S	The band only affects the side channel (stereo width).	Orange

The 0 button below each column bypasses that band individually. All parameters can be automated from your DAW.

Controlling the bands

In the graph: drag a band node horizontally for frequency and vertically for gain. Scroll the mouse wheel over a node to adjust Q (up = narrower). In the columns: drag the knobs vertically for fine adjustment, or double-click a knob and type an exact value - 1.5k is read as 1500 Hz.

4. Dynamic EQ

Every band can be made dynamic: right-click a band node to open a floating panel with four knobs, sized to match the main Freq/Gain/Q knobs below the graph and positioned so it opens clear of the curve around the node you are editing - you can watch the curve change live while you adjust.

Parameter	Function
Threshold (-60...0 dB)	The level in the band's frequency range at which the dynamics start acting.
Range (+-18 dB)	How far the band's gain moves when the level exceeds the threshold. 0 = ordinary static band. Negative range ducks the band when it gets loud (frequency-selective compression); positive range boosts instead.
Attack (0.1-100 ms)	How fast the band reacts when the level rises. Fast attack clamps immediately (can dull transients); slower attack lets the hit through and only tames the ringing tail.
Release (20-1000 ms)	How long the gain change hangs on after the level drops. Match it to the resonance's ring-out time: too short flutters, too long pumps.

Detection is per channel: a band-pass detector at the band's frequency and Q feeds an envelope follower with a soft knee. The knee width scales with the band's Q rather than a fixed 12 dB: about 4 dB (decisive) at high Q, up to 18 dB (gentle) at low Q, since narrow tones sit above masking at a smaller excess than broadband energy does. Bands in MS mode get independent mid and side dynamics.

Active dynamic bands show a ghost ring in the graph - the band's actual gain in real time (green = mid, orange = side), connected to the static node by a thin line. You literally watch the compression work. The hover readout shows the DYN settings, and all dynamics parameters are automatable and included in presets and A/B/C/D.

TIP: A narrow band with gain 0, range -6 and threshold set so the ghost ring only dips on the loudest hits is the classic de-harsh/de-resonance setup: the band is inaudible until it is needed.

5. Band audition (Ctrl+hover)

Hold Ctrl and hover the mouse over a band node in the graph (or its column below) to audition that band in isolation. MSEQ 8 inserts a temporary bandpass filter around the band's own frequency and Q, with a short crossfade in and out, so you hear exactly the range that band is sitting on before you decide to cut or boost it. A thin vertical marker highlights the band's edges in the graph while you audition.

Release Ctrl, or move the mouse off the band, to return to normal listening. Auditioning does not change any parameter, is not automatable, and does not interact with MONITOR or DELTA - it is purely a listening aid.

TIP: Audition a band before setting its dynamics threshold, so you know exactly which frequencies you are about to gate.

6. High-pass and low-pass filters

In addition to the eight bands there is a Butterworth high-pass (HP) and low-pass (LP) filter. They appear as square nodes labelled HP and LP on the 0 dB line.

Action	Result
Drag the node	Changes the filter's cutoff frequency.
Double-click the node	Toggles the filter on/off. A disabled filter is dimmed.
Right-click the node	Menu (opens anchored at the node): on/off, slope (12/24/48 dB per octave), routing (Stereo / Mid / Side), and Independent Mid/Side.
Hover the node	Readout showing frequency, slope and routing.

Independent Mid/Side mode

Turn on Independent Mid/Side from the right-click menu to split a single HP or LP filter into two nodes - one for mid (green), one for side (orange) - each with its own frequency and slope. This is the cleanest way to high-pass the side channel harder than the centre, so the low end becomes mono and solid without touching the mid channel's bass at all. Turn independent mode off again to collapse back to a single shared node at the mid node's frequency.

TIP: HP in Side mode (or the side node in independent mode) around 100-150 Hz is a classic mastering trick: the low end becomes mono and solid without losing width higher up.

7. The graph and the curves

The x axis is logarithmic 20 Hz-20 kHz with markings (50, 100, 200, 500, 1k, 2k, 5k, 10k). The y axis shows dB with markings along the left edge. Two EQ curves are always drawn: green for the mid channel's combined response and orange for the side channel's, including the HP/LP contributions.

Zooming the dB scale

Scroll the mouse wheel over the y-axis zone (left edge) to zoom the scale between +-6 and +-30 dB. The zoom is purely visual - the parameter range (+-18 dB) is unchanged.

Crosshair with frequency and note

As the mouse moves across the graph, a vertical line follows it, and a label fixed in the top-left corner of the plot shows frequency, nearest note (A4 = 440 Hz) and deviation in cents, e.g. 437 Hz - A4 -12 cent - a fixed position was chosen so the readout never covers the MID/SIDE/RES legend row or the Find Resonances chips. Perfect for identifying the pitch of a resonance.

Hover readout

Hover a node to see all of the band's values (number, frequency, gain, Q, M/S mode) in a box next to the node, while the matching band column below the graph is highlighted.

8. The spectrum analyser

The background shows the signal's frequency content in real time, measured after the EQ (but before output gain), calibrated to true dBFS so the curve never clips against the top of the display. Pull a band down and you will immediately see the content drop. The analysis uses a 4096-point FFT with 75% overlap (about 43 updates per second), plus a +3 dB/octave display tilt that makes music look flat instead of sloping downwards.

Two readings are taken from the same FFT block: the drawn curve uses a power/RMS average per point for a smooth, readable trace, while resonance detection (RES/Find Resonances) uses the peak value per point so it stays sensitive to narrow, high-Q peaks that an average would blur away. On top of that, a light frequency-domain smoothing (a small triangular window over neighbouring points, similar to fractional-octave smoothing) removes visual raggedness on the curve without touching the SPEED control's timing. The display runs at up to 30 fps with frame rate-independent smoothing (lower when the window is unfocused - see chapter 2), and the SPEED settings are real time constants (release about 0.12 / 0.3 / 0.8 s).

The legend (top left of the graph)

Control	Function
MID	Shows/hides the mid spectrum (green filled area).
SIDE	Shows/hides the side spectrum (orange filled area).
ST	Shows/hides the stereo spectrum: the total energy sqrt(mid^2+side^2) (purple-grey). Off by default.
RES	Toggles resonance detection markers (see section 9). Detection itself always runs in the background, so Find Resonances has signal to work with even if RES is off.
SPEED	Cycles the analyser's response time: FAST / MED / SLOW. SLOW gives a calm, readable picture of tonal content; FAST tracks transients. Stored with the project.
FREEZE	Freezes a snapshot of the visible spectra as thin reference outlines, e.g. to compare before/after an adjustment. Click again to release.
FIND RESONANCES	Starts the resonance assistant (see section 10): listens for 8 seconds and proposes dynamic bands for the worst resonances.

9. Resonance detection

With RES active, the plugin looks for narrow peaks that persistently (about one second) rise roughly 4 dB above their spectral neighbourhood - typical room or instrument resonances. The three strongest per channel are marked with a vertical line reaching down to the 0 dB line and a frequency/note label printed beside each marked node - green for mid, orange for side. When two markers land close together in frequency, their labels stack into separate rows instead of overlapping, so you can always read both.

Detection runs on the post-EQ spectrum, which makes the workflow self-correcting: place a narrow cut band (high Q) on a marked frequency, pull down until the marker goes out, and the next strongest resonance takes its place in the top list. Built-in hysteresis keeps the markers from flickering between neighbouring peaks. The detector itself runs continuously in the background regardless of the RES toggle, which only controls whether the markers are drawn.

How the detector decides what is a problem

The detection chain models the ear in five steps. It analyses a dedicated RMS spectrum (about 400 ms energy average, independent of the display's SPEED setting), so it sees sustained energy rather than transient spray. The spectral neighbourhood is scaled to the ear's critical bands (ERB): wide in the bass, narrower in the mids, using a median so nearby resonances do not hide each other. A frequency-stability gate rejects wandering content (vocal formants, vibrato) - only stationary peaks qualify. A masking gate drops peaks that sit below the masking threshold of nearby energy and would be inaudible anyway. Finally, peaks with a strong sub-harmonic at f/2, f/3 or f/4 are penalised as probable musical overtones rather than problems.

Hearing-weighted ranking

On top of this, the detector is perceptually weighted with a rough inverted equal-loudness curve (about 80 phon): the presence region gets a +6 dB priority centred around 3.5 kHz where the ear is most sensitive, while the low end (-12 dB towards 20 Hz) and the extreme top (-6 dB towards 20 kHz) are de-prioritised. A modest 3.5 kHz peak therefore outranks a bigger but far less annoying 80 Hz bump - the markers point at what actually hurts.

> *TIP: Use the crosshair to read the note of a resonance - if the same pitch recurs in several octaves it is often the room or the instrument's fundamental.*

Mono-compatibility warning

The same detector data is also used to watch for excessive low-frequency energy in the side channel relative to the mid channel - a common cause of phase cancellation when a mix is folded down to mono (phone speakers, club systems, some broadcast chains). When the side channel's low end persistently outweighs the mid's by a wide margin, a warning appears in the graph. It uses hysteresis so it does not flicker on borderline material, and clears itself once the balance settles back down.

> *TIP: If you see the mono-compatibility warning often, high-pass the side channel (section 6) rather than cutting broadband width - it fixes the mono collapse without narrowing the mix.*

10. The Find Resonances assistant

Find Resonances turns the resonance detector into a co-pilot. Click FIND RESONANCES in the legend while representative music is playing. The plugin listens for 8 seconds (a countdown is shown), gathers statistics on which resonances persist, and then presents up to six proposals (always leaving at least two of the eight bands free for manual work) drawn directly in the graph: a ring at the frequency, an arrow showing the suggested cut, and a label with frequency, Q, depth and channel (M/S).

Each proposal is tailored from measurements, with the bandwidth walk first lightly frequency-smoothed so a single noisy bin cannot make a genuinely moderate resonance measure as artificially narrow: Q from the resonance's measured -3 dB bandwidth (clamped 2-10, matching the band parameter's own range), cut depth at about 60% of how far the peak rises above its neighbourhood (2-9 dB - never the full excess, which would sound dead), and threshold set just below the resonance's own level with a Q-scaled margin - tighter (about 2 dB) for narrow, high-Q tones so they are caught consistently, looser (up to 5 dB) for broad, low-Q resonances so musical broadband material is not choked. Attack takes the larger of two periods of the frequency and the detector filter's own settling time (tau ~ Q/(pi*f)), so narrow resonances get a longer attack and transients still pass unharmed. Release comes from the resonator's physical ring-out time (t60 ~ 2.2*Q/f) so the gain follows the decay without flutter or pumping. When two distinct narrow resonances sit close in frequency, both can become separate proposals - several narrow filters are usually kinder to the mix than one wide cut that also removes sound you want to keep.

Button	Action
APPLY	Writes the proposals as dynamic bands to free bands only - bands you have already set are never touched. Mid resonances get M routing, side resonances S.
DISCARD	Throws the proposals away.
UNDO	Appears after APPLY; restores the exact state from before, one click. Dismissed automatically 10 seconds after Apply, or immediately if you click anywhere else in the plugin - either way this only affects the button's visibility, never the applied bands themselves.

If nothing is playing you get NO SIGNAL; if the material is clean, NO PROBLEM RESONANCES FOUND. The proposals are a starting point - audition with DELTA and adjust to taste.

TIP: Run Find Resonances on the loudest, densest section of the song (often the chorus) - that is where resonances build up. Do not run it with DELTA active.

11. Levels, monitoring, delta and bypass

IN/OUT meters

The panel to the right of the graph shows peak level into and out of the plugin with fast attack and smooth release, against a dB scale from 0 to -60. The bar shifts towards orange near 0 dBFS (above -3 dB).

Output gain

The GAIN knob in the header adjusts the output level +-12 dB after all filtering. Use it for level compensation: if you boost several bands, pull the output down until IN and OUT match - otherwise the louder level tricks your ear into thinking everything got better. Changes are ramped so automation is click-free. The gain change is deliberately not shown in the spectrum analyser.

Monitor: hear what you see

The MONITOR selector in the header chooses what you listen to: ST is the normal stereo output, M plays only the mid signal in both ears, and S plays only the side signal (mono in both ears, the easiest way to listen analytically to the width). Switching is click-free (about 10 ms crossfade). An active solo is shown with a green or orange badge in the graph, and the spectra and resonance detection keep showing both channels regardless. The monitor mode is an automatable parameter - remember to return it to ST before rendering.

> *TIP: Solo the side channel while adjusting S-mode bands: you hear exactly the signal the orange curve is filtering.*

Delta: hear what you are removing

The DELTA button next to MONITOR plays the difference between the dry and the processed signal - exactly what the EQ is taking away. It is the fastest way to verify that a dynamic band is catching only the resonance and not the music around it. Delta is computed in the M/S domain before the monitor stage, so combining DELTA with MONITOR M or S lets you hear what is being removed from the centre or the width alone. Switching is click-free; an orange DELTA badge appears in the graph. Note that the spectrum shows what you hear, i.e. the delta signal while active - and remember to switch DELTA off before rendering.

Bypass

The BYPASS button bypasses the entire plugin. Active bypass is shown with an orange button and a BYPASSED badge in the top-right corner of the graph. Note that a flat EQ (all gains at 0) sounds identical to bypass - that is expected.

12. Undo and redo

Every change you make - EQ parameters, dynamics, Find Resonances proposals, preset switches - is tracked in a 5-step undo history. Press Ctrl+Z to undo the last change, and Ctrl+Shift+Z or Ctrl+Y to redo.

Undo/redo works on the whole plugin state at once rather than a single parameter, so it is a reliable safety net for experimenting freely: try a proposal, tweak a band, switch a preset - if it does not work out, a few presses of Ctrl+Z gets you back. Rapid, continuous changes such as dragging a knob are grouped into a single undo step once you stop moving it, rather than one step per pixel of movement.

> *TIP: Undo/redo and the A/B/C/D snapshots solve different problems: undo steps back through recent history, while A/B/C/D hold whole alternative versions you can compare at will. Use undo to recover from a mistake, and A/B/C/D to compare deliberate alternatives.*

13. Presets and A/B/C/D comparison

Factory presets

Preset	Description
Default	All bands zeroed, HP/LP off.

Preset	Description
Vocal Clarity	HP at 80 Hz, dip in the boxy low mids, presence boost in mid, air on side.
Wide Master	HP on the side channel at 120 Hz (mono bass), treble lift on the sides for width.
Low-End Focus	Bass boost in the mid channel, slight scoop just above.

User presets

Choose Save preset... at the bottom of the preset menu, give your setting a name and save. The preset is stored as an XML file in your user folder (AppData/Roaming/MSEQ8/Presets on Windows) and appears in the menu below the factory presets. The files can be backed up or shared between computers.

A/B/C/D snapshots

The A-D buttons in the header hold four independent snapshots of every parameter. Click a letter to switch there; your changes are automatically stored in the slot you leave. Use them to compare different EQ ideas on the same material before committing. Snapshots are the session's working memory - to keep a result permanently, save it as a preset.

14. Quick reference -- mouse and keyboard

Action	Where	Effect
Drag node	Band node in the graph	Frequency (horizontal) and gain (vertical)
Drag node	HP/LP node	Cutoff frequency
Mouse wheel	Over a band node	Q (up = narrower)
Right-click	Band node	Opens the dynamics panel directly: Threshold, Range, Attack, Release
Ctrl + hover	Band node or column	Audition that band's frequency range in isolation
Mouse wheel	Over the y-axis zone (left edge)	Zooms the dB scale +-6...+-30
Double-click	HP/LP node	Filter on/off
Double-click	Knob (Freq/Gain/Q/Output)	Type an exact value (1.5k = 1500)
Right-click	HP/LP node	Menu (anchored at the node): on/off, slope, routing, Independent Mid/Side
Hover	Node	Readout + highlighted band column
Click	Legend item	Toggle MID/SIDE/ST/RES/FREEZE, cycle SPEED, start FIND RESONANCES
Click	APPLY / DISCARD / UNDO chips	Accept, reject or revert Find Resonances proposals

Action	Where	Effect
Ctrl+Z	Anywhere in the plugin	Undo the last change (up to 5 steps)
Ctrl+Shift+Z or Ctrl+Y	Anywhere in the plugin	Redo
Drag	Bottom-right corner	Resize the window (min. 1020x550)

Automatable parameters

All sound-shaping parameters are exposed to DAW automation: per band Freq, Gain, Q, M/S mode, bypass, Dyn Threshold, Dyn Range, Dyn Attack and Dyn Release; for HP/LP on/off, frequency, slope, routing and Independent Mid/Side; plus Output Gain, Monitor, Delta and global bypass. Parameter changes are smoothed (~20 ms), so fast automation is click-free. Display choices (spectra, SPEED, RES, zoom, window size) are UI settings and are not automatable, but they are stored with the project.

15. Suggested workflows

Cleaning up a stereo mix

1. Run Find Resonances on the chorus and APPLY the proposals - they arrive as dynamic bands with tailored Q, depth, attack and release. 2. Audition with DELTA: you should hear mostly ringing tones, not music; adjust range/threshold where needed. 3. Enable RES and handle anything the assistant left behind with manual narrow bands, using Ctrl+hover to confirm what each candidate band is sitting on first. 4. Enable HP in Side mode (or Independent Mid/Side) around 100-150 Hz for a solid mono low end. 5. Level-compensate with GAIN until IN and OUT match, then compare with BYPASS. If a step does not work out, Ctrl+Z is always one keystroke away.

Widening a mix without losing focus

Put bands 6-8 in S mode and add 1-3 dB between 5-15 kHz. Solo the side channel with MONITOR S to hear exactly what you are shaping, and watch the ST spectrum to confirm the width grows where you want it. Save as A, make an alternative version in B, and switch between them with your ears - not your eyes.

Taming harshness only when it appears

Follow the crosshair to the harsh region (often 2.5-5 kHz), Ctrl+hover to confirm the range by ear, then place a band there with gain 0 and Q around 3, right-click it and set range to about -6 dB with the threshold adjusted until the ghost ring dips only on the loudest, harshest moments. The mix keeps its energy; only the painful peaks are caught.